E0X CSTR Energy Balance Example: Excel part

Decision Guide: How Do I Change Reactor Temperature?

When using the $G(T)$-$R(T)$ method, there are two primary ways to modify reactor operation:

Option 1: Change the Feed Temperature (T_0)

Ask yourself: Do I want to shift the R(T) line left or right?

If YES: Change T0

Increase T0

Increase Feed Temperature
↓
Shift R(T) Right
↓
Higher Operating Temperature

Decrease T0

Decrease Feed Temperature
↓
Shift R(T) Left
↓
Lower Operating Temperature

Key Idea

Changing T0
↓
Shifts R(T) without significantly changing its slope.

Option 2: Change the Heat Exchanger (UA)

Ask yourself: Do I want to change how aggressively heat is removed?

If YES: Change UA

Increase UA

Better Heat Removal
↓
Steeper R(T)
↓
Cooler Reactor

Decrease UA

Less Heat Removal
↓
Flatter R(T)
↓
Hotter Reactor

Key Idea

Changing UA
↓
Rotates R(T) by changing its slope.

Quick Troubleshooting Guide

Reactor Temperature Too High?

Try: Lower T0 or Increase UA

Reactor Temperature Too Low?

Try: Increase T0 or Decrease UA

Want Better Temperature Control?

Increase UA. A larger UA allows the heat exchanger to remove heat more effectively.

Most Important Rule

Feed Temperature
↓
Moves R(T)

Heat Exchanger (UA)

↓
Rotates R(T)

When analyzing non-isothermal CSTRs, this simple rule provides a quick way to predict how operating conditions will change the reactor's steady-state temperature.